

## **CURRICULUM VITAE**

**MAHENDRA PRATAP YADAV**

**Assistant Professor**

**Department of Computer Science & Engineering  
Indian Institute of Information Technology, Pune,  
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### **OBJECTIVE:**

Looking forward with my organization where there is an opportunity to share, contribute and upgrade my knowledge for development of self and the organization.

### **CURRENT POSITION:**

- Presently working as an Assistant Professor in Department of Computer Science and Engineering at **Indian Institute of Information Technology Pune (IIIT Pune), Maharashtra – 411041 (01 December-2022 to till).**

### **AREA OF RESEARCH:**

- **Cloud Computing, Artificial Intelligence Machine Learning, Formal Method**

### **PROFESSIONAL QUALIFICATION:**

- **Ph. D.**  
**Thesis Title:** Optimal Resource Provisioning for Containerized Applications  
**Specialization:** Cloud Computing  
**Department:** Computer Science and Engineering  
**Institute Name:** Motilal Nehru National Institute of Technology Allahabad, Prayagraj, India  
**Registration Date:** 12-01-2016  
**Award Date:** 09-09-2021
- **M. Tech.**  
**Thesis Title:** An Interface Formalism of Web Service Based Protocol Specification and Component Adapter Using Pi-Calculus.  
**Specialization:** Information Security  
**Department:** Computer Science and Engineering  
**Institute Name:** Motilal Nehru National Institute of Technology Allahabad, Prayagraj, India  
**CPI:** 8.125 (81.20%)  
**Session:** 2010-2012
- **B. Tech. in Information Technology** from U.P. Technical University Lucknow (U.P) with 67.04% in the session: 2002-2006.

#### **OTHER QUALIFICATION:**

- Qualified **UGC-NET-2018**
- Qualified **GATE-2008**

#### **ACADEMIC PROFILE:**

- Intermediate passed in 1999 from U.P. Board Allahabad (U.P.).
- High School passed in 1997 from U.P. Board Allahabad (U.P.).

#### **TEACHING EXPERIENCE: (Total-06 Years)**

- Worked as an Assistant Professor in the department of Computing Science & Engineering at SRM University, NCR Campus Modinagar, Ghaziabad, UP – 201204 **(01 July-2022 to 30-11-2022)**.
- Worked as an Assistant Professor (Grade-II) in School of Computing Science & Engineering at VIT University Bhopal, Madhya Pradesh – 466114 **(02-August-2021 to 17-June-2022)**.
- Worked as a Guest faculty in the Department of Computer Science and Engineering at Motilal Nehru National Institute of Technology Allahabad, Prayagraj, India **(30-July-2015 to 16-December-2015)**.
- Worked as a Visiting faculty in the Department of Computer Science and Engineering at Motilal Nehru National Institute of Technology Allahabad, Prayagraj, India **(21-January-2015 to 31-May-2015)**.
- Worked as Assistant Professor in the Department of Computer Science and Engineering at College of Engineering & Rural Technology Meerut (U.P.) India **(28-August-2012 to 30- April-2014)**.
- Worked as Lecturer in the Department of Computer Science and Engineering at College of Engg. & Rural Technology Meerut (U.P.), India **(21-August-2007 to 24- July-2010)**.
- Worked as Lecturer in the Department of Information Technology at Bhagwant Institute of Technology Muzaffarnagar (U.P.), India **(01-February-2007 to 20- August-2007)**.

#### **LIST OF ADMINISTRATIVE POSITIONS HELD:**

- Chairperson (Physical Stock Verification Committee) (Financial Year 2023-24)
- Faculty In charge (National Education Policy (NEP-2020)) (From 09th February 2024 to Continue)
- Warden- Boys Hostel (From 03<sup>rd</sup> October 2023 to 02 September 2024)
- Nodal Officer Visvesvaraya PHD Scheme (From 13<sup>th</sup> September 2023 to Continue)
- Faculty In-charge Hindi Rajbhasha Committee (From 02<sup>nd</sup> June 2023 to Continue)

#### **AREA OF INTREST:**

- Computer Programming, Data structure, Theory of Automaton, Design analysis of algorithm

## SOFTWARE SKILLS:

- **LANGUAGES:** C, C++, Java.
- **DATABASE:** SQL.

## RESEARCH PAPERS (COMMUNICATED /ACCEPTED/ PUBLISHED) IN JOURNALS (SCIE):

- Subhash Yadav, Shubham Shukla, Ritu Tiwari, and **Mahendra Pratap Yadav**, “Path planning for mobile robots with waypoint management using a hybrid nature inspired optimization algorithm” published in Cluster Computing (2025) DOI<https://doi.org/10.1007/s10586-025-05111-8>
- Padma Nimbhore, Tanmoy Hazra, Ritu Tiwari, and **Mahendra Pratap Yadav**, “Classification of Cotton Crop Disease using Hybrid Model and MDfC feature extraction method”, accepted for publication in the Journal of Phytopathology.
- Sanjeev Sharma, **Mahendra Pratap Yadav**, “A collective approach to reach known and unknown target in multi agent environment using nature inspired algorithms”, published in Cluster Computing (2024). <https://doi.org/10.1007/s10586-024-04523-2>
- Rupesh Kumar Dewang, **Mahendra Pratap Yadav**, Surbhit Awasthia, Om Raja, Arvind Mewadaa,, Kamlakant Laxman Bawankulea “Data Secure Application: An Application that Allows Developers to Store User Data Securely using Blockchain and IPFS” published in Multimedia Tools and Applications Volume 83, pages 45491–45517 (2023).
- Brajesh Kumar Umrao, **Mahendra Pratap Yadav** and Dharmendra Kumar Yadav. "Adaptive model for the placement of network services" published in Journal of Information Science and Engineering 40, DOI: 10.6688/JISE.202401\_40 (1).0010 ().
- Nisha Pal, **Mahendra Pratap Yadav** and Dharmendra Kumar Yadav, “Appropriate Number of Analogues in Analogy Based Software Effort Estimation using Quality Datasets” published in Journal of Cluster Computing, publisher Springer, (Indexing - SCIE).
- **Mahendra Pratap Yadav**, Rohit and Dharmendra Kumar Yadav, “**Resource Provisioning Through Machine Learning in Cloud Services**” published 24 July 2021 in Arabian Journal for Science and Engineering Vol. No 47, 1483–1505 (2022) (ISSN: 2191-4281) publisher Springer, <https://doi.org/10.1007/s13369-021-05864-5>, (Indexing - SCIE).
- Mahendra Pratap Yadav, Rohit and Dharmendra Kumar Yadav, “**Maintaining the Container Sustainability through Machine Learning**” published 22 July 2021 in Journal of Cluster Computing (2021) Vol. No.- 24(4) page no- 3725–3750, publisher Springer, <https://doi.org/10.1007/s10586-021-03359-4>, ISSN: 1573-7543, (Indexing - SCIE).
- **Mahendra Pratap Yadav**, Nisha Pal and Dharmendra Kumar Yadav, “**A Formal Approach for Docker Container Deployment**” published 25 May 2021, in Journal of Concurrency and Computation: Practice and Experience (2021) publisher Wiley, Volume 33, Issue 20, <https://doi.org/10.1002/cpe.6364>, ISSN:1532-0634, (Indexing - SCIE).
- **Mahendra Pratap Yadav**, Nisha Pal and Dharmendra Kumar Yadav “**Resource Provisioning for Containerized Applications**”, published 13 May 2021 in Journal of Cluster Computing(2021) Vol. No.- 24(4) page no-2819–2840, publisher Springer <https://doi.org/10.1007/s10586-021-03293-5>, ISSN: 1573-7543, (Indexing - SCIE).

## RESEARCH PAPERS (COMMUNICATED /ACCEPTED/ PUBLISHED) IN JOURNALS (SCOPUS):

- **Mahendra Pratap Yadav**, Nisha Pal and Dharmendra Kumar Yadav, “**Verification of Cloud System**

**Elasticity using BigMC**” published in Journal of System Assurance Engineering and Management, publisher Springer Vol. 13, Page No. 2208–2220 (2022). <https://doi.org/10.1007/s13198-022-01627-z> (Indexing -Scopus).

- Nidhi Sindhwani, Rohit Anand, Meivel S., Rati Shukla, **Mahendra Pratap Yadav**, Vikash Yadav, **“Performance Analysis of Deep Neural Networks Using Computer Vision”**, published 14-October 2021 in EAI Endorsed Transactions on Industrial Networks and Intelligent Systems, Europe (ISSN: 2410-0218), <http://dx.doi.org/10.4108/eai.13-10-2021.171318>, (Indexing - Scopus).
- **Mahendra Pratap Yadav**, Gaurav Raj, Harishchandra Akarte, and Dharmendra Kumar Yadav, **“Horizontal Scaling for Containerized Application Using Hybrid Approach”** published 31-12-2020 in International journal of Ingénierie des Systèmes d’Information, Vol. 25, No. 6, pp. 709-718. <https://doi.org/10.18280/isi.250601>, ISSN: 1633-1311 (print); 2116-7125 (online) (Indexing - Scopus).
- **Mahendra Pratap Yadav**, Harishchandra Akarte, and Dharmendra Kumar Yadav, **“Container Elasticity: Based on Response Time using Docker”** published 12 October, 2020, in International journal of Recent Advances in Computer Science and Communications. (Formerly: Recent Patents on Computer Science) publisher Bentham science, ISSN (Online): 2666-2566, pp-1-13 doi:10.2174/2666255813999201012192010, (Indexing - Scopus).
- **Mahendra Pratap Yadav**, Akanksha Kunwar, Ram Chandra Bhushan, Dharmendra Kumar Yadav, **“Web Server Workload Prediction using Time Series Model”** accepted in International Journal of Advanced Intelligence Paradigms publisher Inderscience-2020, ISSN online 1755-0394, DOI: 10.1504/ijaip.2022.10034175 (Indexing- Scopus).

#### **PATENT (GRANT/PUBLISHE/COMMUNICATED):**

- Patent publishing on the topic **“Mirror List Query Architecture of Multi Architecture Cloud Platform Mirror Warehouse”** in 2022.

#### **RESEARCH PAPERS (PUBLISHED/ACCEPTED) IN BOOK CHAPTERS:**

- **Mahendra Pratap Yadav**, Harishchandra Akarte, Dharmendra Kumar Yadav, **“Efficient Resources Utilization of Containerized Applications using TOPSIS”** published as a book chapter in book with entitled Recent Developments in Artificial Intelligence & Communication Technologies, publisher Bentham Science in 2022.
- **Mahendra Pratap Yadav**, Harishchandra Akarte, Dharmendra Kumar Yadav, **“An Approach for Load Balancing Through Genetic Algorithm”** published as a book chapter in book with entitled ”Smart and Sustainable Intelligent Systems” publisher Scrivener Publishing (Wiley) in March 2021, pp-511-524, ISBN: 978-1-119-75058-1, <https://doi.org/10.1002/9781119752134.ch36>.

#### **RESEARCH PAPERS (PUBLISHED/ACCEPTED) IN CONFERENCES:**

- Nisha Pal, **Mahendra Pratap Yadav** and Dharmendra Kumar Yadav, **“Modeling and Verification of Web Services Composition Using CWB-NC Tool”** published (22 June 2021) in the " INCET 2021:2<sup>nd</sup> International Conference of Emerging Technologies” ISBN:978-1-7281-7029-9 organize on date 21th - 23th May, 2021 at Jain College of Engineering & Technology, Belgaum Karnataka, INDIA.
- **Mahendra Pratap Yadav**, Nisha Pal, and Dharmendra Kumar Yadav, **“Workload Prediction over Cloud Server using Time Series Data”** published (15 march, 2021) in the "Confluence-2021:11<sup>th</sup>

International Conference on Cloud Computing, Data Science & Engineering (28th - 29th January, 2021), ISBN:978-1-6654-1451-7 at Amity University Uttar Pradesh, Noida, India.

- **Mahendra Pratap Yadav**, and Dharmendra Kumar Yadav “**Workload Prediction for Cloud Resource Provisioning using Time Series Data**”, presented in 10th International Conference on Soft Computing for Problem Solving - SocProS 2020. Springer, Singapore, December 18-20, 2020, Organized by Indian Institute of Technology Indore, 14 October 2021, vol 139, pages 447-459, ISBN-978-981-16-2712-5.
- **Mahendra Pratap Yadav**, Mhd Feeroz, Vinod Kumar Yadav, “**Mining the Customer Behavior using Web Usage Mining in E-commerce**”, published (31 December 2012) in (ICCCNT-2012) Third International Conference on Computing Communication and Networking Technology, July 26 -28, 2012, INSPEC Accession Number: 13252164, organized by CCET Tamilnadu, India.
- Vinod Kumar Yadav, **Mahendra Pratap Yadav**, and Dharmendra Kumar Yadav, “**Reliable Task Allocation in Heterogeneous Distributed System with Random Node Failure: Load Sharing Approach**”, published (24 December 2012) in TURING100 - INTERNATIONAL CONFERENCE ON COMPUTING SCIENCES (ICCS-2012), 14-15 September-2012, ISBN- 978-1-4673-2647-6, organized by Lovely Professional University Punjab, India.
- **Mahendra Pratap Yadav**; Pankaj Keserwani; Shefali Gosh Samundar; “**An Efficient Web Mining Algorithm for Web Log Analysis: E-Web Miner**”, , published (07 May 2012) in first International Conference on Recent Advance in Information Technology, 15-17 March 2012, ISBN:978-1-4577-0697-4, Organized by IIT (ISM) Dhanbad, India.
- C.P. Sahu,; **Mahendra Pratap Yadav**,; Manoj Rana.; Ritesh Srivastava; “**A Simple Delayed Acknowledgement Scheme**”, published (18 November 2010) in International Conference on Computer and Communication Technology (ICCCT), 17-19 Sept. 2010 Digital Object Identifier: 10.1109/ICCCT.2010.5640509 Publication Year: 2010 , Page(s): 393 – 396, ISBN:978-1-4244-9034-9, organized by MNNIT Allahabad, India.
- **Mahendra Pratap Yadav**, Mhd Feeroz, and D. K, Yadav “**Mining the Customer Behavior using asociation rule in E-commerce**” published in National conference Jigyasa 2012 Organized by BIT M. Nagar(U.P.).

#### KEYNOTE SPEAKER/ SESSION CHAIRED:

- Delivered a talk on “**DevOps: Software Development and Operations**” at Dr. Babasaheb Ambedkar Technological University, Lonere, Maharashtra (2023).
- Delivered a talk on “**Artificial Intelligence and it’s Real Life Applications**” at ISBM College of Engineering, Pune Maharashtra (2023).
- Session the chaired in “**International Conference On Intelligent Systems and Smart Infrastructure (ICISSI-2022)**” which is jointly organized by Shambhunath Institute of Engineering and Technology, Prayagraj (U.P.) INDIA, Institute Of Engineering and Technology, Lucknow (U.P.) INDIA and Manipal University Jaipur, India
- Delivered a talk (as a Keynote speaker) on “**Container Based Virtualization: A Modern Approach to Deliver the Cloud Services**” in 16<sup>th</sup> International Conference for Internet Technology and Secured Transactions (ICITST-2021).

### CONFERENCES ATTENDED:

- Attended a tutorial on “**Network Security and WiMAX**” in 2010 IEEE International Conference on Computer and Communication Technology (ICCCT), 2010 organized by MNNIT Allahabad.
- Attended a tutorial on “**Cloud Computing and security**” in 2011 IEEE International Conference on Computer and Communication Technology (ICCCT), 2011 organized by MNNIT Allahabad.

### FDPs/ WORKSHOPS/SEMINARS ATTENDED:

- One week **AICTE Training and Learning (ATL) Faculty Development Program** on “**Artificial Intelligence**”, from 10-12-2019 to 14-12-2019, at MNNIT Allahabad, India.
- One week **Workshop** on “**Current Trends in Cyber Security (CTC-2018)**”, from 27-01-2018 to 31-01-2018, organized by department of computer science and engineering at MNNIT Allahabad, India.
- One week **Faculty Development Program** on “**Cloud Computing with AWS**” from 07-08-2017 to 13-08-2017, organized by E&ICT Academy, IIT Guwahati held at KKR & KSR Institute of Technology & Science, Guntur, Andhra Pradesh, India.
- One week **Workshop** on “**Data Analytics and Machine Learning with R**” from 29-05-2017 to 02-06-2017, organized by department of computer science and engineering at MNNIT Allahabad, India.
- One week **Workshop** on “**Network Simulation (WNS-2016)**” from 28-09-2016 to 02-10-2016, organized by department of computer science and engineering at MNNIT Allahabad, India.
- Two weeks (One week online (from 27-04-2015 to 24-05-2015) and one week physical participation (from 25-05-2015 to 30-05-2015 )) **ISTE ITTP Workshop** on “**Introduction to Design of Algorithm**” conducted by IIT Kharagpur at remote center MNNIT Allahabad, India.
- Two weeks **Faculty Development Program** on “**Curriculum Enhancement Awareness in Information in Information Security**” (**SOOCHNA SUREKSHA SAMIKSHA-2009**) from 14-12-2009 to 24-12-2009, organized by department of computer science and engineering at MNNIT Allahabad, India and Sponsored by AICTE.

### STRENGTHS:

- Good communication and interpersonal skills.
- Ability to work with co-operation.
- Leadership quality

**PERSONAL PROFILE:**

**Name** : Mahendra Pratap Yadav  
**Father's Name** : Sri. Ram Pher Yadav  
**Date of birth** : 05<sup>th</sup> Sept 1982.  
**Gender** : Male.  
**Nationality** : Indian.  
**Religion** : Hindu.  
**Languages known** : English & Hindi.  
**Hobbies** : Watching Cricket, Playing Chess, Listening Songs, Reading Book.

**Declaration:**

I hereby declare that the information furnished above is true to the best of my knowledge.

Place:

Date: (MAHENDRA PRATAP YADAV)

Note\*: Relevant documents will be produced on demand.